because the parts of that immovable space in which those motions are performed do by no means come under the observations of our senses.[4]

So Newton leaves us in a somewhat awkward position. He puts absolute space front and center in the description of the most basic and essential element of physics—motion—but he leaves its definition vague and acknowledges his own discomfort about placing such an important egg in such an elusive basket. Many others have shared this discomfort.

Space Jam

Einstein once said that if someone uses words like "red," "hard," or "disappointed," we all basically know what is meant. But as for the word "space," "whose relation with psychological experience is less direct, there exists a far-reaching uncertainty of interpretation."[5] This uncertainty reaches far back: the struggle to come to grips with the meaning of space is an ancient one. Democritus, Epicurus, Lucretius, Pythagoras, Plato, Aristotle, and many of their followers through the ages wrestled in one way or another with the meaning of "space." Is there a difference between space and matter? Does space have an existence independent of the presence of material objects? Is there such a thing as empty space? Are space and matter mutually exclusive? Is space finite or infinite?

For millennia, the philosophical parsings of space often arose in tandem with theological inquiries. God, according to some, is omnipresent, an idea that gives space a divine character. This line of reasoning was advanced by Henry More, a seventeenth-century theologian/philosopher who, some think, may have been one of Newton's mentors.[6] He believed that if space were empty it would not exist, but he also argued that this is an irrelevant observation because, even when devoid of material objects, space is filled with spirit, so it is *never* truly empty. Newton himself took on a version of this idea, allowing space to be filled by "spiritual substance" as well as material substance, but he was careful to add that such spiritual stuff "can be no obstacle to the motion of matter; no more than if nothing were in its way."[7] Absolute space, Newton declared, is the sensorium of God.